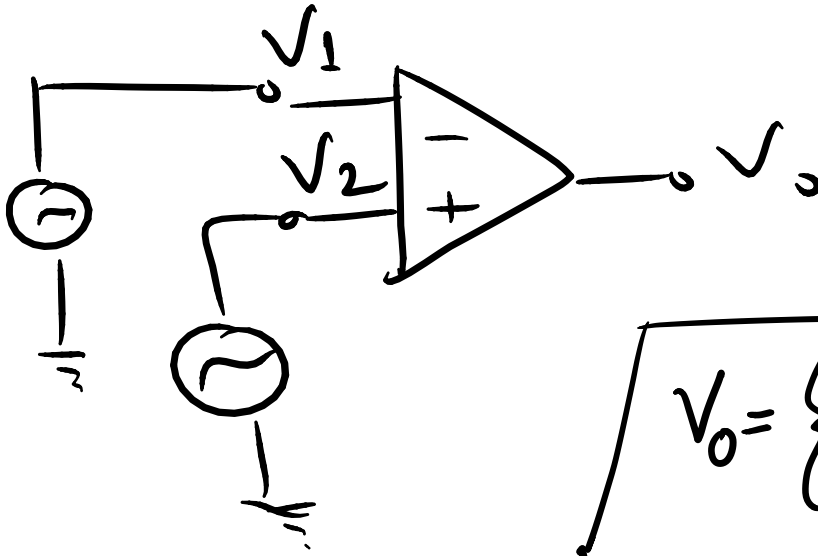


Enstrumantasyon Devreleri:

* Karşılaştırıcı

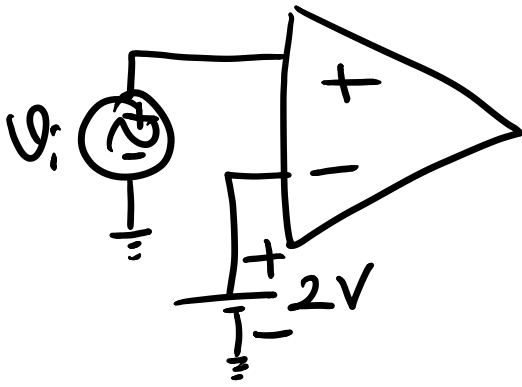


$$V_0 = A_d (V_2 - V_1)$$
$$A_d = 10^5, 10^6$$

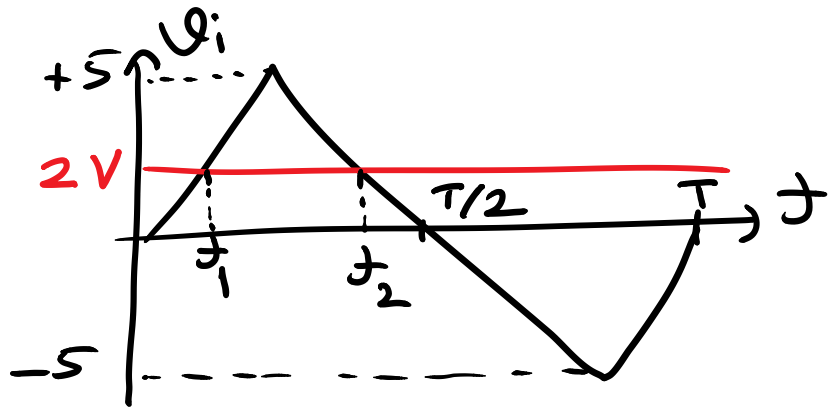
--- gsm

$$V_0 = \begin{cases} +V_{cc} & V_2 > V_1 \\ -V_{cc} & V_1 > V_2 \end{cases}$$

Örnek:



$$\pm V_{cc} = \pm 10V$$

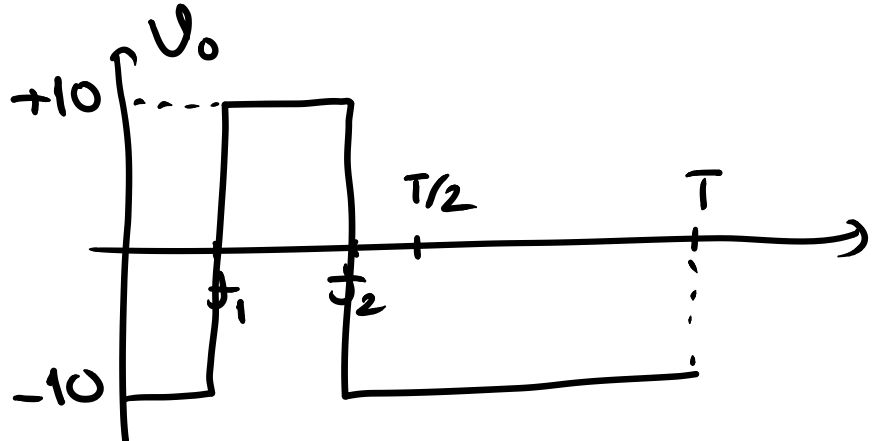


$V_i > 2$ ise

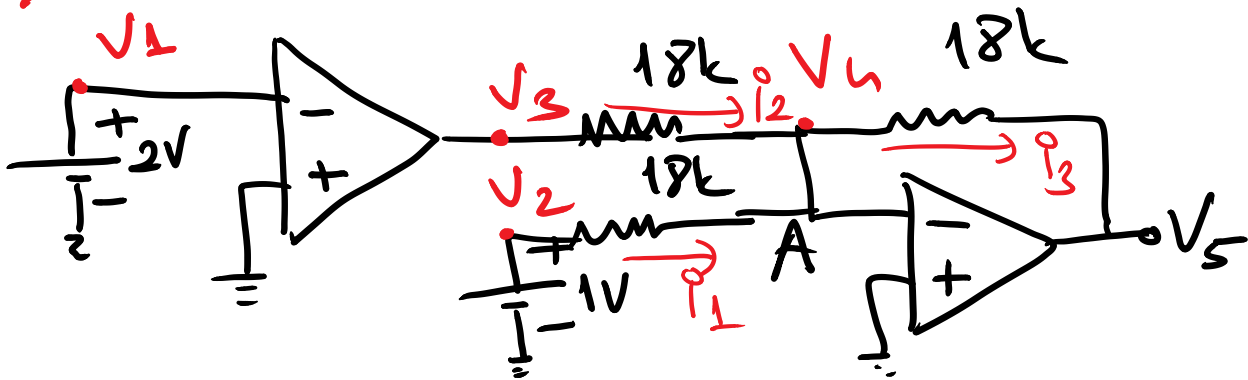
$$V_0 = +V_{cc} \quad (+10V)$$

$V_i < 2$ ise

$$V_0 = -V_{cc} \quad (-10V)$$



Örnek:



$$\pm V_{CC} = \pm 5V$$

$$\boxed{V_1 = 2V}, \quad \boxed{V_2 = 1V}$$

1. op-amp'ta (kararlıdır op-amp)

$$V_- > V_+ \text{ old. için } \begin{pmatrix} V_- = 2V \\ V_+ = 0V \end{pmatrix}$$

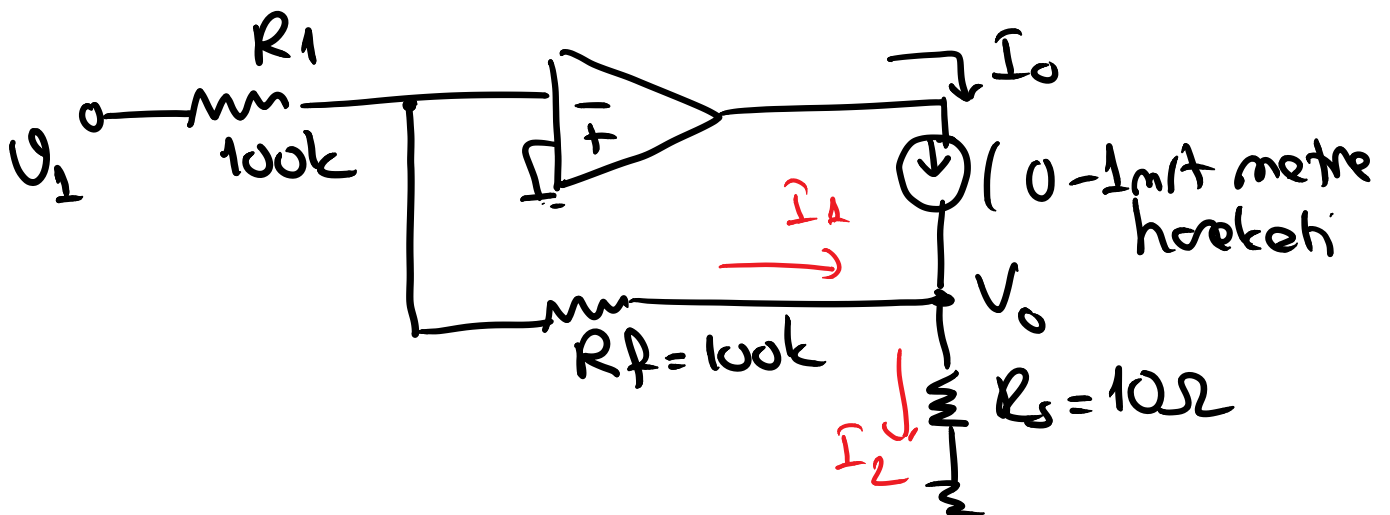
$$\boxed{V_3 = -5V} \text{ } (-V_{CC}) \text{ olur.}$$

A düğümünde:

$$i_1 + i_2 = i_3 \quad \text{ve} \quad V_- = V_+ = 0V.$$

$$\frac{1}{18k} + \frac{-5V}{18k} = \frac{-V_5}{18k} \Rightarrow \boxed{1-5 = -V_5}$$
$$\boxed{V_5 = 4V}$$

DC mili voltmètre:



$$I_0 + I_1 = I_2 \Rightarrow I_0 = I_2 - I_1$$

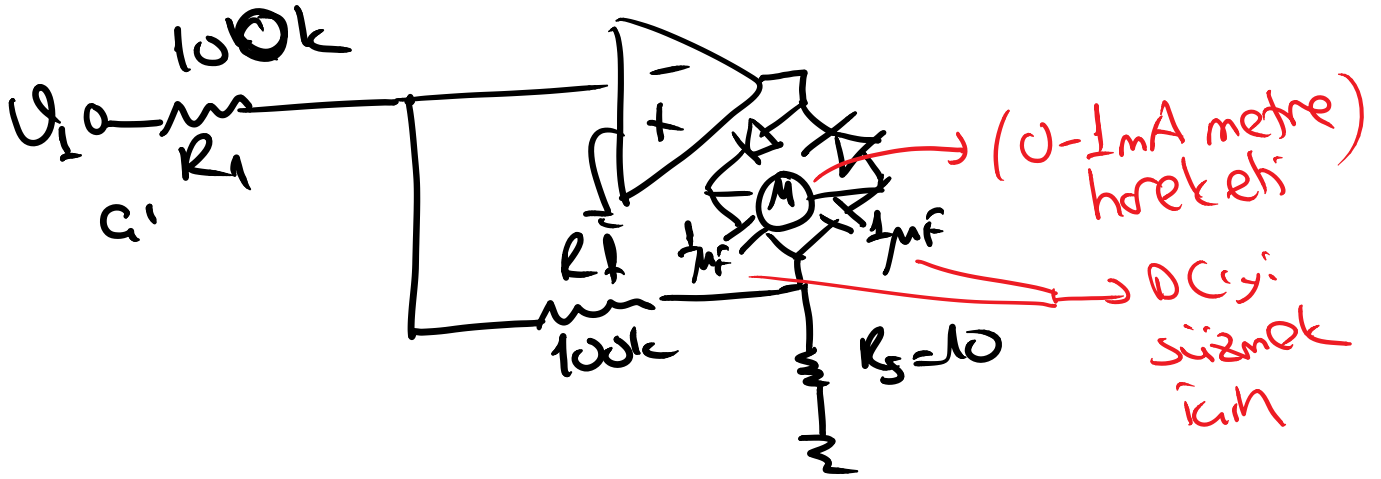
$$I_0 = \frac{V_0}{10} - \frac{0 - V_0}{100k} = \frac{V_0}{10} + \frac{V_0}{100k}$$

71 mal (circled in red)

$$\frac{V_1}{100k} = -\frac{V_0}{100k} \Rightarrow V_0 = -V_1$$

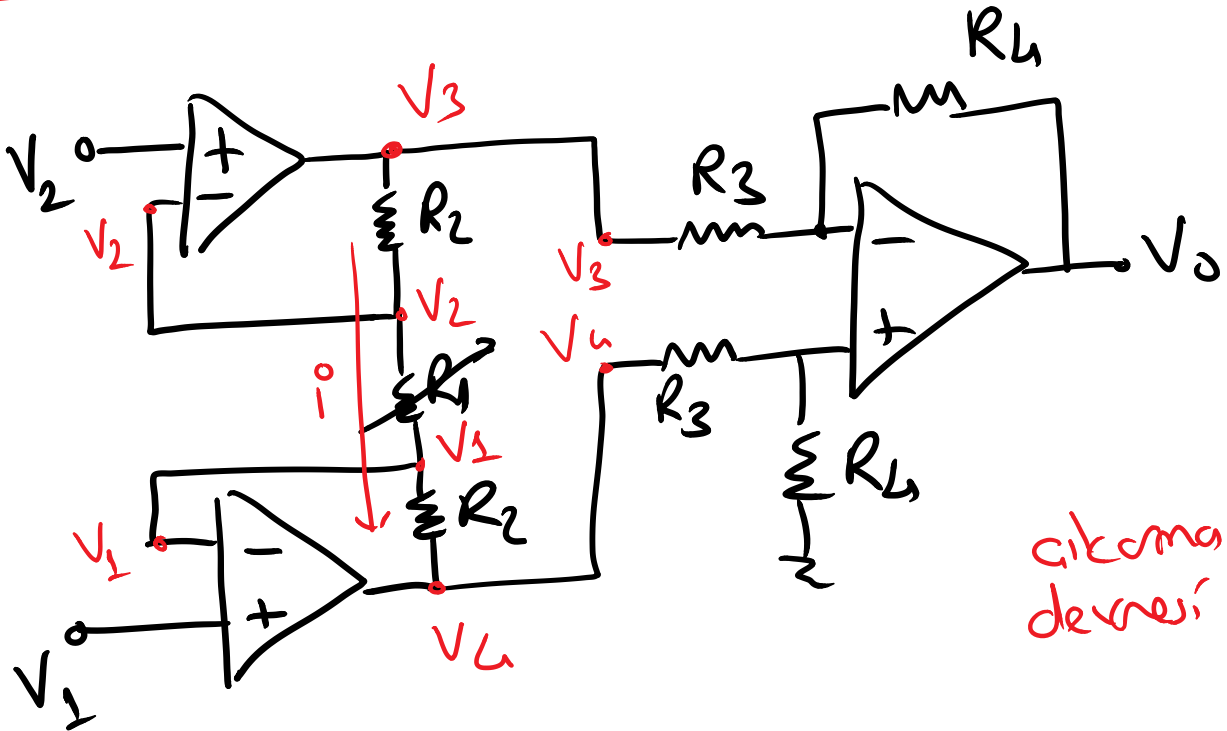
$$I_0 = -\frac{V_1}{10} \Rightarrow \boxed{\frac{I_0}{V_1} = \frac{1}{10}}$$

AC milivoltmetre



$$\left| \frac{I_0}{V_1} \right| = \frac{R_f}{R_1} \cdot \frac{1}{R_s} = \frac{1 \text{ mA}}{10 \text{ mV}}$$

Enstrümantasyon Yüklekçisi:



$$i_0 = \frac{V_3 - V_4}{2R_2 + R_1} = \frac{V_2 - V_1}{R_1}$$

$$V_3 - V_4 = \left(\frac{2R_2 + R_1}{R_1} \right) V_2 - V_1$$

komo deresi:

$$V_0 = \frac{R_4}{R_3} (V_4 - V_3)$$

$$V_0 = \frac{R_4}{R_3} \cdot (V_3 - V_4)$$

$$V_0 = \frac{R_4}{R_3} \cdot \frac{2R_2 + R_1}{R_1} \cdot (V_1 - V_2)$$

